Robust Frequency Synchronization for OFDM-Based Cognitive Radio Systems

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***Abstract*—Cognitive radio employs spectrum sensing to facilitate coexistence of different communication systems over a same frequency band. A peculiar feature of this technology is the possible presence of interference within the signal bandwidth, which considerably complicates the synchronization task. This paper investigates the problem of carrier frequency estimation in an orthogonal frequency-division multiplexing (OFDM)-based cognitive radio system that operates in the presence of narrowband interference (NBI). Synchronization algorithms devised for conventional OFDM transmissions are expected to suffer from significant performance degradation when the received signal is plagued by NBI. To overcome this difficulty, we propose a novel scheme in which the carrier frequency offset (CFO) and interference power on each subcarrier are jointly estimated through maximum likelihood (ML) methods. In doing so we exploit two pilot blocks. The first one is composed of several repeated parts in the time-domain and provides a CFO estimate which may be affected by a certain residual ambiguity. The second block conveys a known pseudo-noise sequence in the frequency-domain and is used to resolve the ambiguity. The performance of the proposed algorithm is assessed by simulation in a scenario inspired by the IEEE 802.11g WLAN system in the presence of a Bluetooth interferer.**

***Index Terms*—Cognitive radio, frequency estimation, interference detection, maximum likelihood estimation.**

I. INTRODUCTION

THE recent proliferation of wireless communication systems has led to an ever increasing demand for frequency spectrum, which has become a very scarce resource. A promising solution for alleviating spectrum limitations is based on the cognitive radio concept [1]. This technology allows frequency reuse and facilitates coexistence of different wireless services on a same frequency band. A natural way to achieve this goal consists of exploring the radio environment and adjusting the transmission parameters so as to reduce interference among simultaneously active systems. The favorite air-interface for cognitive radio is based on orthogonal frequency-division multiplexing (OFDM) due to its inherent flexibility in allocating power and data rate over distinct subchannels. In particular, OFDM can generate non-contiguous groups of subcarriers to fill existing gaps in the frequency spectrum while placing unmodulated subcarriers over subbands occupied by other communication systems. A practical application where cognitive radio can usefully be applied is the recently standardized IEEE 802.11g wireless local area network (WLAN) [2], which adopts an OFDM physical layer and operates in the unlicensed ISM band as the Bluetooth system [3]. The latter is based on a frequency-hopping technology and is viewed by the WLAN as narrowband interference (NBI). In case of collision with an IEEE 802.11g packet, Bluetooth interference can be alleviated by turning off the WLAN interfered subcarriers [4]. Clearly, this operation requires accurate spectrum sensing in order to detect the presence of NBI within the signal bandwidth.

The problem of spectrum sensing in multicarrier transmissions has received some attention in the last few years and references [5]-[8] provide a good sample of the results obtained in this area. In particular, the approach in [8] relies on the repetition of a pilot block in the time domain. Since the two repeated parts remain identical after passing through the channel, their subtraction at the receiver will only leave the contribution of noise plus interference, which is then easily detected. Unfortunately, this method cannot work in the presence of carrier frequency offsets (CFOs) induced by Doppler shifts and/or oscillator instabilities. The reason is that in such a case the received pilot blocks are identical except for a phase shift and, accordingly, some residual signal power will remain after subtraction.

The existence of NBI within the OFDM spectrum considerably complicates the synchronization process [9]. As is known, in multicarrier systems timing errors result in interblock interference (IBI) between adjacent OFDM blocks and must be properly counteracted to avoid severe error rate degradations. In this respect, the use of a sufficiently long guard interval in the form of a cyclic prefix (CP) can provide intrinsic protection against IBI at the expense of some extra overhead. In practical applications, data transmission is organized in frames and training blocks are normally placed at the beginning of each frame to assist the synchronization process. If the training blocks are preceded by large enough CPs, it is likely that a rough timing estimate can be achieved even in the presence of NBI by means of conventional methods based on suitable correlations in the time-domain [10]-[12]. After coarse timing adjustment, the receiver has to perform frequency recovery in order to align its local oscillator to the received carrier frequency. Depending on the application, the CFO may exceed the subcarrier spacing Δf. In such a case, it is customary to divide the CFO into an integer part, multiple of Δf, plus a fractional part. If not properly compensated, the former results into a shift of the subcarrier indices while the latter produces interchannel interference (ICI) due to loss of orthogonality among subcarriers. Conventional methods for estimating the fractional offset operate in the time-domain and measure the

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phase shift between the repetitive parts of a dedicated pilot block [10]-[13]. In contrast, the integer offset is typically estimated in the frequency-domain using an additional pilot block carrying a known pseudo-noise (PN) sequence [10], [14]. These methods provide good results as long as the received signal is only affected by channel distortions and thermal noise, whereas significant degradations are expected in the presence of non-negligible NBI. The maximum likelihood (ML) estimator of the fractional CFO based on the observation of two identical OFDM blocks was given by Moose in [15]. In that work, the CFO is estimated by comparing the phases of corresponding subcarriers in the received blocks. In principle, this scheme can work even in the presence of NBI provided that all interference jammed subcarriers are properly identified and excluded from the estimation process. As mentioned previously, however, NBI detection in the presence of synchronization errors is a challenging task.

In this paper we address the problem of frequency recovery in OFDM transmissions affected by NBI. In doing so we resort to ML methods and assume that interference is Gaussian distributed with zero mean and unknown power [8]. A training block composed by L repeated parts is employed to jointly estimate the CFO and the NBI power on each subcarrier. This results into a frequency estimation scheme which is inherently robust against NBI and with an acquisition range of $\pm L/2$ the subcarrier spacing. The latter can be widened by means of a second training block carrying a known PN sequence. Simulations indicate that the proposed estimator is suited for OFDM-based cognitive radio and its accuracy is close to the relevant Cramer-Rao bound (CRB) even in the presence of strong NBI. A basic assumption behind our scheme is the absence of IBI. As discussed previously, such condition can be achieved provided that the training blocks are preceded by adequately long CPs and a rough timing estimate is available at the receiver. On the other hand, it is desirable that the CP of data blocks is made just greater than the length of the channel impulse response (CIR) in order to minimize the system overhead. This means that accurate timing estimates should be obtained after CFO recovery and NBI detection. Unfortunately, the problem of fine timing estimation in the presence of NBI is not addressed in the literature and remains an issue for future work.

The rest of the paper is organized as follows. Section II describes the system model and introduces basic notation. In Sect. III we discuss the joint ML estimation of the CFO and interference power, while Sect. IV illustrates how to enlarge the frequency acquisition range. Simulation results are presented in Sect. V and some conclusions are drawn in Sect. VI.

II. System model

We consider an OFDM system employing N subcarriers modulated by PSK or QAM data symbols and potentially affected by NBI. To avoid IBI, each OFDM block is preceded by a CP longer than the CIR. At the receiver side the incoming waveform is down-converted to baseband and sampled at a rate $f_s = N\Delta f$, where Δf is the frequency distance between adjacent subcarriers. Due to Doppler shifts and/or oscillator instabilities, the frequency f_{LO} of the local oscillator

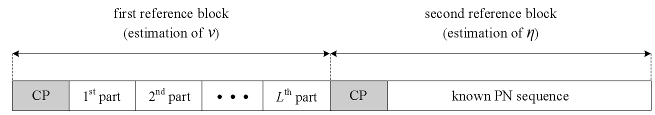

Fig. 1. Pilot blocks for CFO estimation.

employed for signal demodulation is not exactly equal to the received carrier frequency f_c. Denoting by $\xi = (f_c - f_{LO})/\Delta f$ the CFO normalized to the subcarrier distance, we may write the received time-domain samples as

$$x(k) = e^{j2\pi\xi k/N}s(k) + I(k) + n(k) \tag{1}$$

where $s(k)$ is the useful signal component, $n(k)$ is thermal noise and $I(k)$ accounts for possible NBI. If not properly compensated, the CFO destroys orthogonality among subcarriers and results into significant performance degradations. Our goal is to design a CFO estimation algorithm which may work even in the presence of strong NBI. For this purpose, we assume that two pilot blocks are appended in front of each data frame. As depicted in Fig. 1, the first block is divided into L identical parts, each containing $M = N/L$ samples. In practice, it can be generated in the frequency-domain by transmitting pilot symbols on the subcarriers with indices multiple of L while setting the others to zero. The second training block has no repetitive structure and its subcarriers are modulated by a known PN sequence of length N. Note that such a preamble structure is compliant with IEEE 802.11 based WLAN systems, where the frame starts with 10 short pilot blocks which are used for coarse frequency acquisition, followed by two additional blocks for fine frequency synchronization and channel estimation [16].

To proceed further, we decompose ξ into a *fractional part*, less than $L/2$ in magnitude, plus an *integer part* which is multiple of L. The normalized CFO can thus be rewritten as

$$\xi = \nu + \eta L \tag{2}$$

where η is an integer and ν belongs to the interval $(-L/2, L/2]$. We exploit the first block in Fig. 1 to get an estimate of ν while the second block is employed to recover η. Clearly, the estimation of η is unnecessary if the maximum value of $|\xi|$ is guaranteed to be less than $L/2$ since in such a case we have $\eta = 0$.

III. Estimation of the fractional CFO

In this Section we derive a scheme for estimating the fractional part of the CFO. The basic assumption is that coarse timing synchronization has already been established through standard techniques and the CP length is such that the residual timing error does not produce any IBI.

A. Problem formulation

A natural way for localizing and mitigating NBI is to operate in the frequency domain. Following this approach, the received samples belonging to the first pilot block are divided into L consecutive segments, each containing M samples and corresponding to one of the repetitive parts in which the block is divided. Every segment is identified by an index

$\ell = 0, 1, \ldots, L-1$ and converted in the frequency domain through an $M-$point discrete Fourier transform (DFT) operation. We denote by $X_\ell(m)$ the mth DFT output corresponding to the ℓth segment. Since the repetitive parts of the pilot block remain identical after passing through the channel except for a phase shift induced by the CFO, we may write

$$X_\ell(m) = S(m)e^{j2\pi\ell M\xi/N} + W_\ell(m) \qquad 0 \le m \le M-1 \quad (3)$$

where $S(m)$ is the signal component and $W_\ell(m)$ accounts for background noise plus any possible interference arising from collision with other communication systems. It is worth noting that, although the PN sequence modulating the first pilot block is normally known to the receiver, the quantities $\{S(m)\}$ are plagued by channel distortions and CFO-induced ICI and, accordingly, they are treated as unknown parameters. Following [8], we model $W_\ell(m)$ as a circularly symmetric Gaussian random variable with zero mean and unknown variance $\sigma^2(m) = \sigma_n^2 + \sigma_I^2(m)$, where σ_n^2 is the noise power (the same at each DFT output) while $\sigma_I^2(m)$ represents the NBI contribution and depends in general on the frequency index m. Clearly, $\sigma_I^2(m) = 0$ if the mth DFT output is free from interference. To facilitate the discussion, in all subsequent derivations the quantities $\{W_\ell(m)\}$ are assumed to be statistically independent for different values of m and ℓ.

For any given m, we arrange the DFT outputs into a vector $\mathbf{X}(m) = [X_0(m), X_1(m), \ldots, X_{L-1}(m)]^T$, where the superscript $(\cdot)^T$ denotes the transpose operator. Hence, substituting (2) into (3) and recalling that $M = N/L$ yields

$$\mathbf{X}(m) = S(m)\mathbf{u}(\nu) + \mathbf{W}(m) \qquad 0 \le m \le M-1 \quad (4)$$

where $\mathbf{u}(\nu) = [1, e^{j2\pi\nu/L}, \ldots, e^{j2\pi(L-1)\nu/L}]^T$ while $\mathbf{W}(m) = [W_0(m), W_1(m), \ldots, W_{L-1}(m)]^T$ is Gaussian distributed with zero mean and covariance matrix $\sigma^2(m)\mathbf{I}_L$ (we denote $\mathbf{I}_L$ the identity matrix of order L). Our goal is to exploit vectors $\{\mathbf{X}(m)\ ;\ m = 0, 1, \ldots, M-1\}$ for jointly estimating ν, $\mathbf{S} = [S(0), S(1), \ldots, S(M-1)]^T$ and $\boldsymbol{\sigma}^2 = [\sigma^2(0), \sigma^2(1), \ldots, \sigma^2(M-1)]^T$. A solution to this problem is now derived by resorting to ML methods.

B. ML estimation of the fractional CFO

Given the unknown parameters $(\mathbf{S}, \boldsymbol{\sigma}^2, \nu)$, from (4) it turns out that vectors $\{\mathbf{X}(m)\}$ are statistically independent and Gaussian distributed with mean $S(m)\mathbf{u}(\nu)$ and covariance matrix $\sigma^2(m)\mathbf{I}_L$. Accordingly, the log-likelihood function for $(\mathbf{S}, \boldsymbol{\sigma}^2, \nu)$ takes the form

$$\Lambda(\tilde{\mathbf{S}}, \tilde{\boldsymbol{\sigma}}^2, \tilde{\nu}) = -N\ln(\pi) - L\sum_{m=0}^{M-1}\ln[\tilde{\sigma}^2(m)] - \sum_{m=0}^{M-1}\frac{1}{\tilde{\sigma}^2(m)}\left\|\mathbf{X}(m) - \tilde{S}(m)\mathbf{u}(\tilde{\nu})\right\|^2 \quad (5)$$

where $\tilde{\mathbf{S}}$, $\tilde{\boldsymbol{\sigma}}^2$ and $\tilde{\nu}$ are trial values of $\mathbf{S}$, $\boldsymbol{\sigma}^2$ and ν, respectively, while $\|\cdot\|$ represents the Euclidean norm of the enclosed vector. The joint ML estimate of the unknown parameters is the location where $\Lambda(\tilde{\mathbf{S}}, \tilde{\boldsymbol{\sigma}}^2, \tilde{\nu})$ achieves its global maximum. The latter can be found following a three-step procedure. In the first step we keep $\tilde{\nu}$ and $\tilde{\mathbf{S}}$ fixed and let $\tilde{\boldsymbol{\sigma}}^2$ vary. In such a case the maximum of the log-likelihood function is achieved for

$$\hat{\sigma}^2(m; \tilde{\mathbf{S}}, \tilde{\nu}) = \frac{1}{L}\left\|\mathbf{X}(m) - \tilde{S}(m)\mathbf{u}(\tilde{\nu})\right\|^2 \qquad 0 \le m \le M-1. \quad (6)$$

Substituting this result into (5) and skipping irrelevant factors and additive terms independent of $(\tilde{\mathbf{S}}, \tilde{\nu})$ yields

$$\Lambda(\tilde{\mathbf{S}}, \tilde{\nu}) = -\sum_{m=0}^{M-1}\ln\left[\left\|\mathbf{X}(m) - \tilde{S}(m)\mathbf{u}(\tilde{\nu})\right\|^2\right]. \quad (7)$$

In the second step we fix $\tilde{\nu}$ and maximize $\Lambda(\tilde{\mathbf{S}}, \tilde{\nu})$ with respect to $\tilde{\mathbf{S}}$. This produces

$$\hat{S}(m; \tilde{\nu}) = \frac{1}{L}\mathbf{u}^H(\tilde{\nu})\mathbf{X}(m) \qquad 0 \le m \le M-1 \quad (8)$$

where the superscript $(\cdot)^H$ means "Hermitian transpose". Substituting (8) into the right-hand-side of (7) gives the *concentrated* likelihood function of ν

$$\Lambda(\tilde{\nu}) = -\sum_{m=0}^{M-1}\ln\left[\|\mathbf{X}(m)\|^2 - \Phi(m, \tilde{\nu})\right] \quad (9)$$

where $\Phi(m, \tilde{\nu})$ is the periodogram of $\mathbf{X}(m)$, which reads

$$\Phi(m, \tilde{\nu}) = \frac{1}{L}\left|\sum_{\ell=0}^{L-1}X_\ell(m)\ e^{-j2\pi\ell\tilde{\nu}/L}\right|^2. \quad (10)$$

The final step provides the estimate of ν as the location where $\Lambda(\tilde{\nu})$ is maximum, i.e.,

$$\hat{\nu} = \arg\max_{\tilde{\nu}}\{\Lambda(\tilde{\nu})\}. \quad (11)$$

We refer to the above algorithm as the maximum likelihood estimator (MLE) of the fractional CFO.

C. Remarks and practical adjustments

1) From (9) and (10) we see that $\Lambda(\tilde{\nu})$ is periodic of period L and, in consequence, its maxima occur at a distance of L from each other. This means that MLE gives ambiguous estimates unless ν belongs to the interval $|\nu| \le L/2$, which is the estimation range of the MLE. Note that the requirement $|\nu| \le L/2$ is always fulfilled by definition of the fractional CFO.
2) An interesting (albeit unrealistic) situation occurs when *at least* one of the M vectors $\{\mathbf{X}(m)\ ;\ m = 0, 1, \ldots, M-1\}$, say $\mathbf{X}(m')$, is free from noise and interference. In such a case, from (4) we have

$$\mathbf{X}(m') = S(m')\mathbf{u}(\nu) \quad (12)$$

so that the periodogram (10) evaluated at $\tilde{\nu} = \nu$ takes the value

$$\Phi(m', \nu) = L\left|S(m')\right|^2. \quad (13)$$

Substituting (12) and (13) into (9) yields

$$\lim_{\tilde{\nu}\to\nu}\Lambda(\tilde{\nu}) = +\infty \quad (14)$$

which means that MLE provides the true value of ν irrespective of the interference power at the DFT outputs with indices $m \neq m'$. This reveals the remarkable robustness of MLE against NBI.

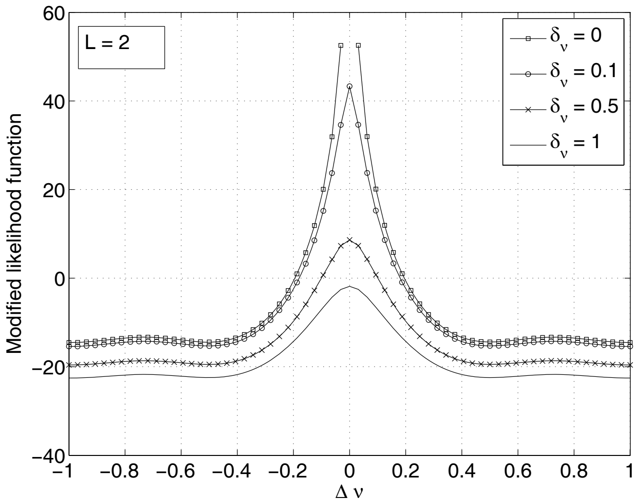

Fig. 2. Modified likelihood function $\overline{\Lambda}(\tilde{\nu})$ vs. $\Delta\nu$ for $L = 2$ and some values of δ_ν.

3) From (14) we expect that the concentrated likelihood function exhibits a very large peak at $\tilde{\nu} = \nu$ whenever some of the vectors $\mathbf{X}(m)$ are nearly free from interference and noise. This fact is advantageous on one hand, but on the other it may cause numerical problems in evaluating $\Lambda(\tilde{\nu})$. A solution to this potential drawback is found by replacing $\Lambda(\tilde{\nu})$ with the following *modified likelihood function*

$$\overline{\Lambda}(\tilde{\nu}) = -\sum_{m=0}^{M-1} \ln\left[\delta_\nu + \|\mathbf{X}(m)\|^2 - \Phi(m,\tilde{\nu})\right] \quad (15)$$

where $\delta_\nu > 0$ is a suitably designed parameter. Applying the Cauchy-Schwartz inequality

$$\left|\sum_{\ell=0}^{L-1} \alpha(\ell)\beta^*(\ell)\right|^2 \leq \sum_{\ell=0}^{L-1} |\alpha(\ell)|^2 \cdot \sum_{\ell=0}^{L-1} |\beta(\ell)|^2 \quad (16)$$

with $\alpha(\ell) = X_\ell(m)$ and $\beta(\ell) = e^{j2\pi\ell\tilde{\nu}/L}$ yields $\Phi(m,\tilde{\nu}) \leq \|\mathbf{X}(m)\|^2$, from which it follows that $\overline{\Lambda}(\tilde{\nu})$ cannot exceed the value $\overline{\Lambda}_{\max} = -M\ln\delta_\nu$. Hence, parameter δ_ν should be chosen such that $\overline{\Lambda}_{\max}$ lies within the dynamic range of the likelihood function calculator. Fig. 2 illustrates a typical realization of $\overline{\Lambda}(\tilde{\nu})$ vs. $\Delta\nu = \tilde{\nu} - \nu$ for some values of δ_ν and $L = 2$ in the absence of noise and interference. As is seen, the amplitude of the peak reduces as δ_ν grows large. Surprisingly, in Sect. V we will show that the use of $\overline{\Lambda}(\tilde{\nu})$ in place of $\Lambda(\tilde{\nu})$ is also advantageous in terms of estimation accuracy. In the ensuing discussion, the estimator employing the metric (15) is called modified MLE (MMLE).

4) Maximizing function $\overline{\Lambda}(\tilde{\nu})$ requires a grid-search over the interval $(-L/2, L/2]$. In practice, the global maximum is sought in two steps. The first one (*coarse search*) calculates $\overline{\Lambda}(\tilde{\nu})$ for a set of $\tilde{\nu}$ values, say $\{\tilde{\nu}_n\}$, covering the uncertainty range of ν and determines the location $\tilde{\nu}_M$ of the maximum over this set. The second step (*fine search*) makes an interpolation between the samples $\overline{\Lambda}(\tilde{\nu}_n)$ and computes the local maximum nearest to $\tilde{\nu}_M$. The coarse search can be efficiently performed using Fast Fourier Transform (FFT) techniques. Specifically, for each $m = 0, 1, \ldots, M-1$ we consider the following zero-padded sequence of length N

$$X'_\ell(m) = \begin{cases} X_\ell(m) & 0 \leq \ell \leq L-1 \\ 0 & L \leq \ell \leq N-1 \end{cases} \quad (17)$$

and compute its FFT at the points

$$\tilde{\nu}_n = \frac{nL}{N}, \qquad -\frac{N}{2} < n \leq \frac{N}{2}. \quad (18)$$

This produces $\Phi(m,\tilde{\nu}_n)$, which is next exploited to evaluate the quantities $\left\{\overline{\Lambda}(\tilde{\nu}_n); -N/2 < n \leq N/2\right\}$ as indicated in (15). The largest $\overline{\Lambda}(\tilde{\nu}_n)$ is eventually sought and this provides the coarse estimate of ν.

5) In the ideal case where all of the variances $\{\sigma^2(m)\}$ are known to the receiver, the concentrated likelihood function of ν reduces to the following weighted sum of periodograms

$$\Gamma(\tilde{\nu}) = \sum_{m=0}^{M-1} \frac{\Phi(m,\tilde{\nu})}{\sigma^2(m)}. \quad (19)$$

In the sequel, the estimator employing the metric (19) is referred to as MLE with *perfect variance knowledge* (MLE-PVK). Since this scheme effectively exploits some side information that is not available to MLE, it is expected to outperform the estimator (11).

6) A suboptimal approach for CFO recovery is to design the frequency estimation algorithm without taking NBI into consideration. This amounts to setting $\sigma^2(m) = \sigma_n^2$ in (19) for $m = 0, 1, \ldots, M-1$, thereby leading to a new metric

$$\Gamma'(\tilde{\nu}) = \sum_{m=0}^{M-1} \Phi(m,\tilde{\nu}). \quad (20)$$

The corresponding estimator is called MLE for an *interference-free scenario* (MLE-IFS). Interestingly, $\Gamma'(\tilde{\nu})$ can be obtained from the MMLE metric in (15) after letting δ_ν go to infinity. To see how this comes about, we make use of the approximated relation $\ln(\delta_\nu + x) \approx \ln\delta_\nu + x/\delta_\nu$, which applies when $\delta_\nu \gg x$. In this way, we may rewrite $\overline{\Lambda}(\tilde{\nu})$ in (15) as

$$\overline{\Lambda}(\tilde{\nu}) = -M\ln\delta_\nu - \frac{1}{\delta_\nu}\sum_{m=0}^{M-1} \|\mathbf{X}(m)\|^2 + \frac{1}{\delta_\nu}\sum_{m=0}^{M-1} \Phi(m,\tilde{\nu}) \quad (21)$$

which reduces to $\Gamma'(\tilde{\nu})$ after neglecting irrelevant factors and additive terms independent of $\tilde{\nu}$.

7) By invoking the asymptotic efficiency property of MLE, the accuracy of the frequency estimate (11) is expected to approach the corresponding CRB for relatively large values of M and L. In the Appendix it is shown that

$$CRB(\nu) = \frac{3\overline{SINR}^{-1}}{2\pi^2 N(1 - 1/L^2)} \quad (22)$$

where

$$\overline{SINR} = \frac{1}{M}\sum_{m=0}^{M-1}\frac{|S(m)|^2}{\sigma^2(m)} \tag{23}$$

is the average signal-to-interference-plus-noise ratio (SINR) at the DFT outputs. Interestingly, the bound (22) also applies to the ideal case where all variances $\{\sigma^2(m)\}$ are perfectly known to the receiver.

IV. Estimation of the Integer CFO

If the normalized CFO is guaranteed to be less than $L/2$ in magnitude, the quantity $\hat{\nu}$ is an estimate of ξ and, in consequence, the second training block in Fig. 1 is unnecessary. Otherwise, ξ is expressed as in (2) and an estimate of the integer frequency offset η must be found. This problem is now addressed using ML methods.

A. Problem formulation

In order to compensate for the fractional offset ν, the received samples belonging to the pilot blocks are first counter-rotated at an angular speed $2\pi\hat{\nu}/N$ and subsequently transformed to the frequency-domain by means of two N-point DFT operations. Let $Y_0(n)$ and $Y_1(n)$ $(0 \le n \le N-1)$ be the DFT outputs corresponding to the first and second block, respectively. In case of perfect compensation of ν, both $Y_0(n)$ and $Y_1(n)$ are free from ICI. However, they will be shifted from their correct position by a quantity ηL due to the uncompensated integer CFO. Denoting by $H(n)$ the channel attenuation at the nth DFT output and assuming $\hat{\nu} \approx \nu$, we may write

$$Y_i(n) = H(n)c_i\left(|n-\eta L|_N\right)e^{j2\pi i\eta L N_T/N} + \overline{W}_i(n) \tag{24}$$

where $|n-\eta L|_N$ is the value $n-\eta L$ reduced to the interval $[0, N-1]$, N_T denotes the total length in sampling periods of the second pilot block (included the CP) and $c_i(n)$ is the pilot symbol transmitted onto the nth subcarrier of the ith block, with $i = 0, 1$. Recalling that the first block is composed by L identical parts, it is $c_0(n) = 0$ when n is not multiple of L. The quantities $\{\overline{W}_i(n)\}$ represent the contribution of interference plus noise and are modeled as circularly symmetric Gaussian random variables with zero mean and unknown variance $\sigma_w^2(n)$. They are assumed to be statistically independent for different values of n and i.

For notational simplicity, we denote $\{d_i(n)\ ; -\infty < n < +\infty\}$ the repetition with period N of the pilot sequence $\{c_i(n)\ ; 0 \le n \le N-1\}$. In such a case, $c_i\left(|n-\eta L|_N\right)$ can be equivalently replaced by $d_i\left(n-\eta L\right)$. Hence, letting $\mathbf{Y}(n) = [Y_0(n), Y_1(n)]^T$, $\mathbf{U}(\eta)$ =diag$\{1, e^{j2\pi\eta L N_T/N}\}$ and $\mathbf{d}(n-\eta L) = \left[d_0\left(n-\eta L\right), d_1\left(n-\eta L\right)\right]^T$, we may rewrite (24) in matrix notation as

$$\mathbf{Y}(n) = H(n)\mathbf{U}(\eta)\mathbf{d}(n-\eta L) + \overline{\mathbf{W}}(n) \qquad n = 0, 1, \ldots, N-1 \tag{25}$$

where $\overline{\mathbf{W}}(n) = [\overline{W}_0(n), \overline{W}_1(n)]^T$ is Gaussian distributed with zero mean and covariance matrix $\sigma_w^2(n)\mathbf{I}_2$. In the following, vectors $\{\mathbf{Y}(n); n = 0, 1, \ldots, N-1\}$ are employed to find the joint ML estimates of η, $\mathbf{H} = [H(0), H(1), \ldots, H(N-1)]^T$ and $\boldsymbol{\sigma}_w^2 = [\sigma_w^2(0), \sigma_w^2(1), \ldots, \sigma_w^2(N-1)]^T$.

B. ML estimation of the integer CFO

The log-likelihood function for $(\mathbf{H}, \boldsymbol{\sigma}_w^2, \eta)$ based on the observation of $\{\mathbf{Y}(n)\}$ is expressed by

$$\Psi(\tilde{\mathbf{H}}, \tilde{\boldsymbol{\sigma}}_w^2, \tilde{\eta}) = -2N\ln(\pi) - 2\sum_{n=0}^{N-1}\ln[\tilde{\sigma}_w^2(n)] - \sum_{n=0}^{N-1}\frac{1}{\tilde{\sigma}_w^2(n)}\left\|\mathbf{Y}(n) - \tilde{H}(n)\mathbf{U}(\tilde{\eta})\mathbf{d}(n-\tilde{\eta}L)\right\|^2 \tag{26}$$

where $\tilde{\mathbf{H}}$, $\tilde{\boldsymbol{\sigma}}_w^2$ and $\tilde{\eta}$ are trial values of the unknown parameters. The location where $\Psi(\tilde{\mathbf{H}}, \tilde{\boldsymbol{\sigma}}_w^2, \tilde{\eta})$ achieves its global maximum gives the joint ML estimate of $(\mathbf{H}, \boldsymbol{\sigma}_w^2, \eta)$. Following the same steps of Sect. III.B, we begin by maximizing the log-likelihood function with respect to the variances $\tilde{\boldsymbol{\sigma}}_w^2$. This produces

$$\hat{\sigma}_w^2(n; \tilde{\mathbf{H}}, \tilde{\eta}) = \frac{1}{2}\left\|\mathbf{Y}(n) - \tilde{H}(n)\mathbf{U}(\tilde{\eta})\mathbf{d}(n-\tilde{\eta}L)\right\|^2 \qquad 0 \le n \le N-1. \tag{27}$$

Substituting this result into (26) and skipping irrelevant factors and additive terms, yields

$$\Psi(\tilde{\mathbf{H}}, \tilde{\eta}) = -\sum_{n=0}^{N-1}\ln\left[\left\|\mathbf{Y}(n) - \tilde{H}(n)\mathbf{U}(\tilde{\eta})\mathbf{d}(n-\tilde{\eta}L)\right\|^2\right]. \tag{28}$$

The next step is the maximization of $\Psi(\tilde{\mathbf{H}}, \tilde{\eta})$ with respect to $\tilde{\mathbf{H}}$ for a given $\tilde{\eta}$. After standard calculations we get

$$\hat{H}(n; \tilde{\eta}) = \frac{\mathbf{d}^H(n-\tilde{\eta}L)\mathbf{U}^H(\tilde{\eta})\mathbf{Y}(n)}{\|\mathbf{d}(n-\tilde{\eta}L)\|^2} \qquad 0 \le n \le N-1 \tag{29}$$

having used the identity $\mathbf{U}^H(\tilde{\eta})\mathbf{U}(\tilde{\eta}) = \mathbf{I}_2$. The concentrated likelihood function for η is eventually obtained after substituting (29) into the right-hand-side of (28). This produces $\Psi(\tilde{\eta})$ given in (30), which, after some manipulations, can be equivalently rewritten as in (31).

At this stage we recall that $c_0(n) = 0$ when n is not multiple of L, and the same occurs with $d_0\left(n-\tilde{\eta}L\right)$. In such a case, the argument of the logarithm in (31) reduces to $|Y_0(n)|^2$ and the corresponding term $\ln\left[|Y_0(n)|^2\right]$ can be neglected as it becomes independent of $\tilde{\eta}$. In conclusion, the ML estimate of η is found to be

$$\hat{\eta} = \arg\max_{|\tilde{\eta}| \le |\eta|_{\max}}\{g(\tilde{\eta})\} \tag{32}$$

where $|\eta|_{\max}$ represents the largest expected value of $|\eta|$, which is determined by the stability of the transmitter and receiver oscillators, and $g(\tilde{\eta})$ is obtained from (31) after retaining the only terms that depend on $\tilde{\eta}$ (i.e., those with indices $n = mL$) as shown in (33).

If all pilot symbols have the same magnitude (as occurs when they are taken from a PSK constellation), the denominator in the right-hand-side of (33) is independent of $\tilde{\eta}$ and function $g(\tilde{\eta})$ further simplifies to (34). The estimator employing the metric (34) is referred to as the maximum likelihood estimator (MLE) of the integer CFO.

$$\Psi(\tilde{\eta}) = -\sum_{n=0}^{N-1} \ln \left[\left\| \mathbf{Y}(n) - \frac{\mathbf{U}(\tilde{\eta})\mathbf{d}(n-\tilde{\eta}L)\mathbf{d}^H(n-\tilde{\eta}L)\mathbf{U}^H(\tilde{\eta})\mathbf{Y}(n)}{\|\mathbf{d}(n-\tilde{\eta}L)\|^2} \right\|^2 \right] \tag{30}$$

$$\Psi(\tilde{\eta}) = -\sum_{n=0}^{N-1} \ln \left[\frac{\left| d_0\left(n-\tilde{\eta}L\right) Y_1(n) e^{-j2\pi\tilde{\eta}LN_T/N} - d_1\left(n-\tilde{\eta}L\right) Y_0(n) \right|^2}{\left|d_0\left(n-\tilde{\eta}L\right)\right|^2 + \left|d_1\left(n-\tilde{\eta}L\right)\right|^2} \right] \tag{31}$$

$$g(\tilde{\eta}) = -\sum_{m=0}^{M-1} \ln \left[\frac{\left| d_0(mL-\tilde{\eta}L) Y_1(mL) e^{-j2\pi\tilde{\eta}LN_T/N} - d_1(mL-\tilde{\eta}L) Y_0(mL) \right|^2}{\left|d_0(mL-\tilde{\eta}L)\right|^2 + \left|d_1(mL-\tilde{\eta}L)\right|^2} \right] \tag{33}$$

$$g(\tilde{\eta}) = -\sum_{m=0}^{M-1} \ln \left[\left| d_0(mL-\tilde{\eta}L) Y_1(mL) e^{-j2\pi\tilde{\eta}LN_T/N} - d_1(mL-\tilde{\eta}L) Y_0(mL) \right|^2 \right] \tag{34}$$

C. Remarks

1) Let us temporarily assume that $\overline{\mathbf{W}}(n) = \mathbf{0}$ for some values of n. Then, after substituting (24) into (34) we see that $g(\tilde{\eta})$ approaches infinity for $\tilde{\eta} = \eta$. Although this fact emphasizes a remarkable robustness of the proposed algorithm to NBI, it may cause numerical problems in evaluating $g(\tilde{\eta})$. A solution is found by adopting the same approach of Sect. III.C, which amounts to replacing $g(\tilde{\eta})$ with the modified metric $\overline{g}(\tilde{\eta})$ given in (35). The parameter $\delta_\eta > 0$ in (35) is designed so that the maximum value of $\overline{g}(\tilde{\eta})$, which is $\overline{g}_{\max} = -M \ln \delta_\eta$, does not exceed the dynamic range of the metric calculator. We denote modified MLE (MMLE) the estimator that computes $\hat{\eta}$ looking for the location where $\overline{g}(\tilde{\eta})$ is maximum.
2) Once the quantities $\hat{\nu}$ and $\hat{\eta}$ have been computed, an estimate of the normalized CFO is obtained from (2) as
$$\hat{\xi} = \hat{\nu} + \hat{\eta}L. \tag{36}$$
Clearly, $\hat{\xi}$ will have the same accuracy of $\hat{\nu}$ as long as the integer CFO is perfectly estimated, i.e., $\hat{\eta} = \eta$.
3) If all variances $\{\sigma_w^2(n)\}$ were perfectly known to the receiver, the ML estimate of η would be found by looking for the maximum of the metric $\gamma(\tilde{\eta})$ given in (37). This estimator, which is called MLE-PVK, is expected to outperform MLE due to the increased amount of information available for estimating η.
4) Putting $\sigma_w^2(mL) = \sigma_n^2$ in (37) for $m = 0, 1, \ldots, M-1$ and skipping an irrelevant factor leads to the new metric $\gamma'(\tilde{\eta})$ expressed in (38). This estimator does not take the possible presence of NBI into account and, for this reason, is denoted MLE-IFS. Using the same methods of Sect. III.C, it can be shown that MLE-IFS originates from MMLE as δ_η approaches infinity. Also, it is worth noting that the metric (38) is the same employed by Morelli, D'Andrea and Mengali (MDM) in [14] when no virtual carriers are placed in the second training block.

V. Simulation Results

Computer simulations have been run to assess the performance of the frequency recovery schemes illustrated in the previous sections. The simulation model is inspired by the specifications of the IEEE 802.11g WLAN system and is summarized as follows.

A. Simulation model

The considered WLAN system has $N = 64$ subcarriers and operates in the 2.4 GHz frequency band. The signal bandwidth is 20 MHz, corresponding to a subcarrier distance of 312.5 kHz. The sampling period is $T_s = 50$ ns, so that the useful part of each OFDM block has length 3.2 μs. A CP of 0.8 μs is adopted to eliminate IBI. The discrete-time CIR is composed by 8 channel taps collected into a vector $\mathbf{h} = [h(0), h(1), \ldots, h(7)]^T$. The taps are modeled as independent circularly symmetric Gaussian random variables with zero-mean (Rayleigh fading) and an exponential power delay profile

$$E\{|h(k)|^2\} = \lambda \cdot \exp(-k) \qquad k = 0, 1, \ldots, 7 \tag{39}$$

where the constant λ is chosen such that the channel power is normalized to unity, i.e., $E\{\|\mathbf{h}\|^2\} = 1$. A channel snapshot is generated at each new simulation run and kept fixed over an entire frame.

In addition to background noise with variance σ_n^2, the OFDM signal is affected by a Bluetooth interferer which occupies a bandwidth of 1 MHz and adds Gaussian noise of variance σ_I^2 to 4 contiguous WLAN subcarriers. We define the signal-to-noise ratio (SNR) as $10\text{Log}(\sigma_s^2/\sigma_n^2)$ where $\sigma_s^2 = E\left\{|s(k)|^2\right\}$ is the average power of the received signal component, while the signal-to-interference ratio (SIR) is given by $10\text{Log}(\sigma_s^2/\sigma_I^2)$. Clearly, the SIR is defined only over

$$\overline{g}(\tilde{\eta}) = -\sum_{m=0}^{M-1} \ln \left[\delta_\eta + \left| d_0(mL-\tilde{\eta}L) Y_1(mL) e^{-j2\pi\tilde{\eta}LN_T/N} - d_1(mL-\tilde{\eta}L) Y_0(mL) \right|^2 \right] \tag{35}$$

$$\gamma(\tilde{\eta}) = \sum_{m=0}^{M-1} \frac{\Re e\left\{Y_1(mL)Y_0^*(mL)d_0(mL-\tilde{\eta}L)d_1^*(mL-\tilde{\eta}L)e^{-j2\pi\tilde{\eta}LN_T/N}\right\}}{\sigma_w^2(mL)} \tag{37}$$

$$\gamma'(\tilde{\eta}) = \sum_{m=0}^{M-1} \Re e\left\{Y_1(mL)Y_0^*(mL)d_0(mL-\tilde{\eta}L)d_1^*(mL-\tilde{\eta}L)e^{-j2\pi\tilde{\eta}LN_T/N}\right\} \tag{38}$$

the interference-jammed subcarriers, while the SNR is defined over the entire WLAN bandwidth. The pilot symbols belong to a PSK constellation and the position of the Bluetooth interferer in the WLAN spectrum is randomly varied at each simulation run. To find the fractional CFO, a parabolic interpolation is chosen in the implementation of the fine search.

The accuracy of the fractional CFO estimators is measured in terms of mean square estimation error (MSEE), which is defined as $E\{|\hat{\nu}-\nu|^2\}$. The probability of failure $P_f = \Pr\{\hat{\eta} \neq \eta\}$ is adopted as a performance indicator for integer CFO recovery.

B. Performance assessment for fractional CFO estimation

We begin by assessing the impact of parameter δ_ν on the accuracy of MMLE. Fig. 3 illustrates the MSEE vs. δ_ν for $L = 2$ and 4. Marks indicate simulation results while solid lines are drawn to ease the reading of the graphs. The fractional CFO varies at each simulation run according to a uniform distribution over the interval $(-L/2, L/2]$. The SNR is kept fixed to 15 dB while the SIR over the interfered subcarriers is either -3 or -10 dB. From the discussion in Sect. III.C, we recall that MMLE converges to MLE-IFS for large values of δ_ν, while letting $\delta_\nu = 0$ provides the performance of MLE. Since in Fig. 3 the accuracy of the CFO estimates for small values of δ_ν is virtually independent of the SIR level, we argue that MLE is highly robust to NBI. In contrast, large differences between SIR= -3 dB and SIR= -10 dB are observed as δ_ν increases, meaning that MLE-IFS is severely affected by NBI. As we can see, for each L there exists an optimum value of δ_ν minimizing the MSEE. This fact indicates that MMLE outperforms MLE when δ_ν is properly chosen and may appear surprising at first sight. A possible explanation is that MLE turns out to be the minimum variance unbiased estimator (MVUE) only for large data records (or asymptotically) so that, in principle, better schemes may exist when the estimator operates with a limited observation window. Furthermore, from Fig. 2 we see that $\overline{\Lambda}(\tilde{\nu})$ exhibits a large peak at $\tilde{\nu} = \nu$ if δ_ν approaches zero. This reduces the accuracy of the fine search since the parabolic approximation of $\overline{\Lambda}(\tilde{\nu})$ in the vicinity of the peak becomes less and less accurate. On the other hand, choosing a too large value of δ_ν is not recommended as in such a case MMLE boils down to MLE-IFS, thereby leading to a larger MSEE. Inspection of Fig. 3 reveals that $\delta_\nu = 0.10$ represents a good choice with both $L = 2$ and 4. This value is adopted for MMLE in the subsequent simulations.

Fig. 4 compares the performance of the fractional CFO estimators in terms of MSEE vs. SNR for $L = 2$. The SIR is fixed to -10 dB and the CRB is also shown as a benchmark.

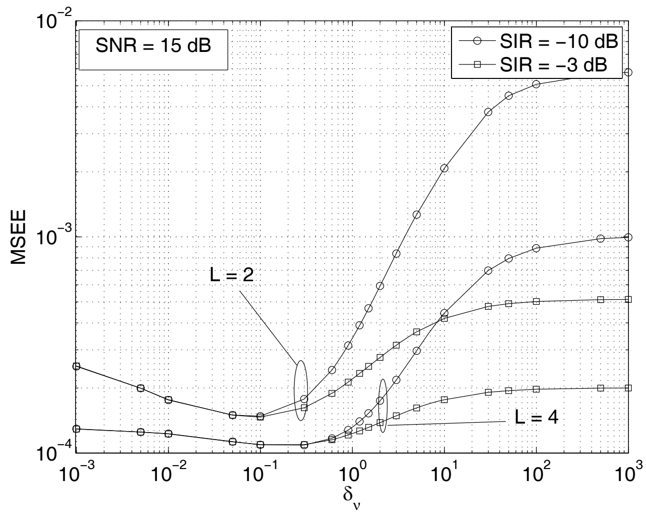

Fig. 3. Accuracy of MMLE vs. δ_ν for SNR= 15 dB and $L = 2$ or 4.

The latter is computed from (22) by numerically averaging the right-hand-side of (23) with respect to the channel realizations. The curve labeled MM illustrates the performance of the frequency estimator proposed by Morelli and Mengali in [13]. This scheme computes the CFO estimate by measuring the phase shift between the L repetitive parts of the training block and reduces to the Schmidl & Cox algorithm [10] when $L = 2$. We see that MMLE outperforms MLE by approximately 3 dB for SNR > 10 dB. As expected, the best accuracy is obtained with MLE-PVK, which approaches the CRB at large SNR values. In contrast, significant degradations are observed with MM and MLE-IFS, which do not take NBI into account.

The results of Fig. 5 have been obtained in the same simulation set-up of Fig. 4, except for the value of L which is now set to 4. We see that the gain of MMLE over MLE is reduced to nearly 1 dB and the accuracy of both schemes keeps close to the CRB. Again, the best performance is achieved by MLE-PVK whereas MM and MLE-IFS exhibit large error floors.

Fig. 6 illustrates the MSEE of the considered schemes vs. the SIR so as to compare their interference rejection capability. The SNR is fixed to 15 dB while $L = 2$. The poor performance of MM and MLE-IFS at low SIR values is a clear evidence of the vulnerability of these methods in the presence of interference. In contrast, the accuracy of the other estimators exhibits a weak dependence upon the SIR thanks to their intrinsic robustness to NBI. Results obtained with $L = 4$ are qualitatively similar and are not shown for space limitations.

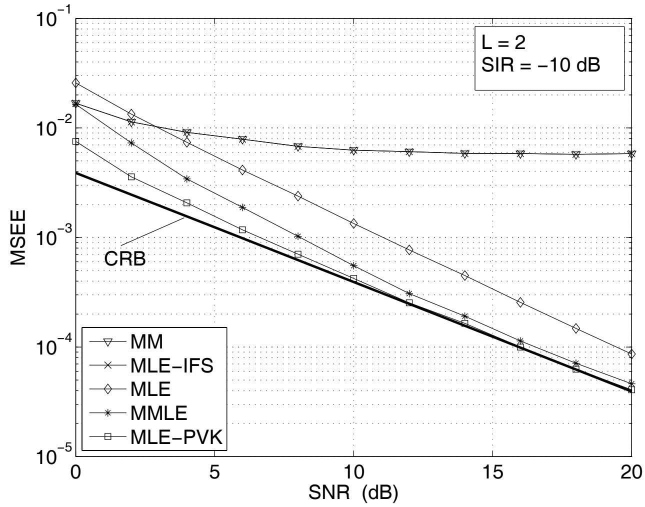

Fig. 4. Accuracy of the frequency estimators vs. SNR for $L = 2$ and SIR $= -10$ dB

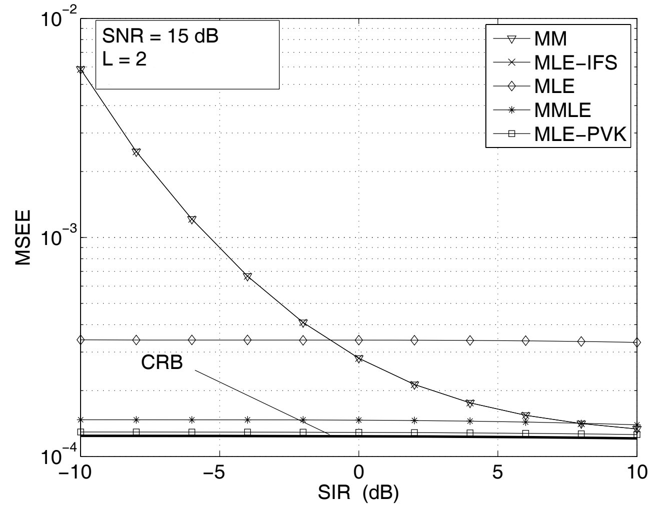

Fig. 6. Accuracy of the frequency estimators vs. SIR for SNR $= 15$ dB and $L = 2$.

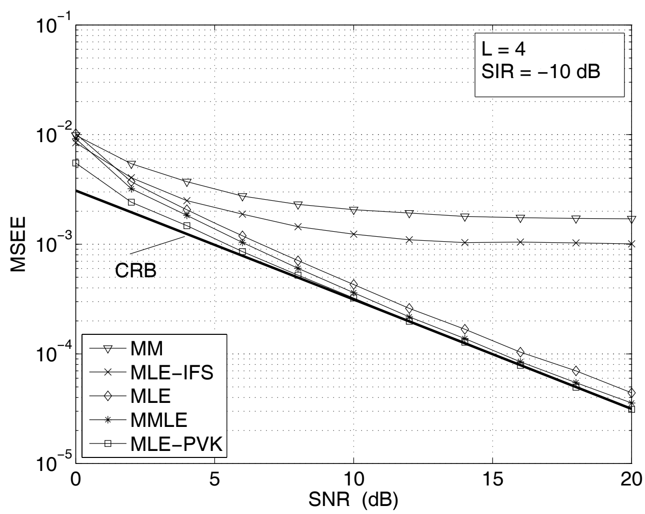

Fig. 5. Accuracy of the frequency estimators vs. SNR for $L = 4$ and SIR $= -10$ dB

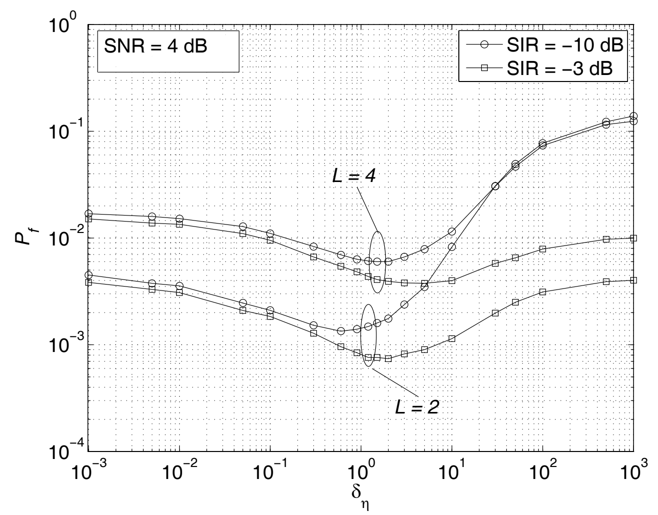

Fig. 7. Probability of failure of MMLE vs. δ_η for SNR $= 4$ dB and $L = 2$ or 4.

C. Performance assessment for integer CFO estimation

As mentioned previously, the presence of a CFO after down-conversion is a consequence of oscillator instabilities and Doppler shifts arising from the relative motion between the transmitter and receiver. Low-cost oscillators have a frequency stability in the order of ten parts per million (ppm) while Doppler shifts are normally limited within a few kilohertz. Thus, recalling that the carrier frequency and subcarrier spacing in the considered WLAN scenario are 2.4 GHz and 312.5 kHz, respectively, the maximum expected normalized CFO is $\xi_{\max} = 0.2$. This means that the integer CFO is always zero and its estimation is thus unnecessary. However, in applications where the subcarriers are closely spaced in frequency, the CFO can be much greater than the subcarrier distance. In such a case, estimating the integer CFO becomes mandatory.

In Fig. 7 we show the probability of failure of MMLE as a function of δ_η for $L = 2$ and 4. Again, letting $\delta_\eta = 0$ provides the performance of MLE, while MLE-IFS is approached for large values of δ_η. The SNR is set to 4 dB while the SIR over the interfered subcarriers is either -3 or -10 dB. The largest expected value of $|\eta|$ is $|\eta|_{\max} = 4$. For simplicity, the fractional CFO has been perfectly compensated, i.e., we let $\hat{\nu} = \nu$. As is seen, the curves are qualitatively similar to those in Fig. 3, indicating that we can minimize P_f by a proper selection of δ_η. In particular, choosing $\delta_\eta = 1$ provides near optimum performance with both $L = 2$ and 4. Recalling that $M = N/L$ with N fixed to 64, we see that the number of additive terms in the metric (35) reduces as L grows large. This explains why in Fig. 7 the best results are obtained with $L = 2$ instead of $L = 4$.

Figs. 8 and 9 illustrate the probability of failure of the considered estimators vs. SNR for $L = 2$ and 4, respectively. The SIR is fixed to -10 dB and MMLE operates with $\delta_\eta = 1$. The curve labeled SC illustrates the performance of the scheme proposed by Schmidl and Cox in [10], where an

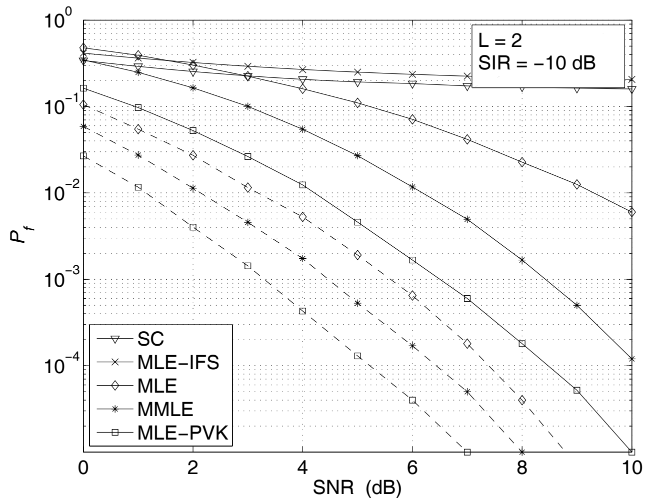

Fig. 8. Probability of failure vs. SNR for $L = 2$ and SIR $= -10$ dB.

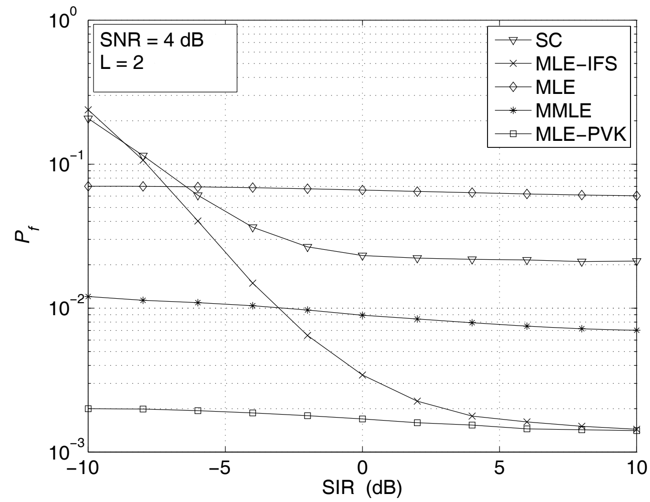

Fig. 10. Probability of failure vs. SIR for $L = 2$ and SNR $= 4$ dB.

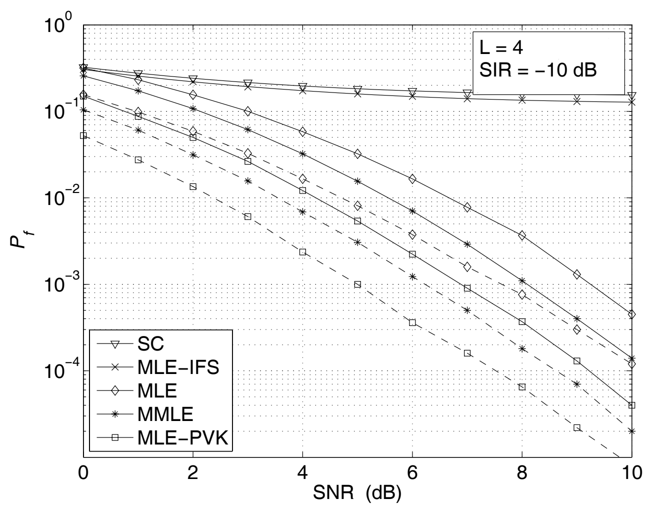

Fig. 9. Probability of failure vs. SNR for $L = 4$ and SIR $= -10$ dB.

estimate of η is found by maximizing the following metric with respect to $\tilde{\eta}$

$$\gamma_{SC}(\tilde{\eta}) = \left| \sum_{m=0}^{M-1} Y_1(mL) Y_0^*(mL) d_0(mL - \tilde{\eta}L) d_1^*(mL - \tilde{\eta}L) \right|^2 . \tag{40}$$

The dashed curves are obtained assuming $\hat{\nu} = \nu$, while solid lines refer to the case where the estimate $\hat{\nu}$ is computed as indicated in Sect. III. Again, MLE-PVK exhibits the best performance while MLE-IFS and SC are plagued by an irreducible error floor due to their poor robustness against NBI. We also see that MMLE outperforms MLE, especially for $L = 2$. The loss due to imperfect compensation of the fractional CFO is remarkable for $L = 2$, but reduces to approximately 2 dB when $L = 4$.

The performance of the considered estimators vs. the SIR is shown in Fig. 10 for SNR $= 4$ dB and $L = 2$. The fractional CFO is compensated using an estimate of ν obtained with the methods discussed in Sect. III. As expected, MLE-IFS and SC perform poorly at low SIR values while the probability of failure of the other schemes depends weakly upon the interference power. Interestingly, when the SIR is relatively large, MLE-IFS and MLE-PVK have similar performance and provide better results then MLE and MMLE. This means that, in the presence of weak NBI, ignoring the interference is preferable than trying to mitigate it.

VI. Conclusions

We have derived a number of synchronization schemes for rapid and robust CFO recovery in OFDM-based cognitive radio systems plagued by narrowband interference. They operate in the frequency domain and exploit two training blocks available for synchronization purposes. The first block is used for estimating the CFO with a certain residual ambiguity while the second block serves to enlarge the frequency acquisition range. In applications where the maximum expected CFO is small enough, no ambiguity occurs and frequency recovery can thus be accomplished upon the receipt of just one pilot block. Interference is modeled as a Gaussian process and its power on each subcarrier is jointly estimated with the CFO using maximum likelihood methods. Comparisons are made with the Cramer-Rao bound and with conventional frequency recovery schemes that do not take NBI into account. Simulations indicate that the proposed methods are capable to remove most of the degradation that NBI imposes on the synchronization process and can be effectively used in WLAN transmissions over unlicensed frequency bands.

Numerical results obtained in the presence of a true Bluetooth interfere lead to similar conclusions and are not shown. In such a case, however, interference will be partially present at all DFT outputs due to the leakage effect, thereby producing an irreducible error floor in the performance curves at high SNR levels.

Appendix

In this Appendix we employ the signal model (4) to derive the CRB. We denote $\mathbf{S}_R$ and $\mathbf{S}_I$ the real and imaginary parts of $\mathbf{S}$, and define $\boldsymbol{\zeta} = (\mathbf{S}_R, \mathbf{S}_I, \boldsymbol{\sigma}^2, \nu)$ the set of unknown parameters. Then, the components of the Fisher information matrix $\mathbf{F}$ are given by [17]

$$[\mathbf{F}]_{i,j} = -E\left\{\frac{\partial^2 \Lambda(\boldsymbol{\zeta})}{\partial \zeta_i \partial \zeta_j}\right\} \qquad 1 \le i,j \le 3M+1 \tag{41}$$

where $\Lambda(\boldsymbol{\zeta})$ is the log-likelihood function in (5), $E\{\cdot\}$ denotes the expectation operator and ζ_ℓ is the ℓth entry of $\boldsymbol{\zeta}$. Substituting (5) into (41), after standard computations we find

$$\mathbf{F} = \begin{bmatrix} \mathbf{D} & \mathbf{v} \\ \mathbf{v}^T & \beta \cdot \mathbf{S}^H \mathbf{P}^{-1} \mathbf{S} \end{bmatrix} \tag{42}$$

where $\mathbf{P}$ and $\mathbf{D}$ are the following diagonal matrices

$$\mathbf{P} = \mathrm{diag}\{\sigma^2(0), \sigma^2(1), \ldots, \sigma^2(M-1)\} \tag{43}$$

$$\mathbf{D} = L \cdot \mathrm{diag}\{2\mathbf{P}^{-1}, 2\mathbf{P}^{-1}, \mathbf{P}^{-2}\} \tag{44}$$

$\mathbf{v}$ is a column-vector of dimension $3M$

$$\mathbf{v} = 2\pi(L-1) \cdot [-\mathbf{S}_I^T \mathbf{P}^{-1} \quad \mathbf{S}_R^T \mathbf{P}^{-1} \quad \mathbf{0}_M^T]^T \tag{45}$$

with $\mathbf{0}_M$ denoting the null vector with M zeros and, finally, $\beta = 4\pi^2(L-1)(2L-1)/(3L)$. Letting $\mathbf{F}^{-1}$ be the inverse of $\mathbf{F}$, the CRB for the estimation of ν is given by

$$CRB(\nu) = \left[\mathbf{F}^{-1}\right]_{3M+1,3M+1}. \tag{46}$$

To proceed further, we define the $(3M+1)-$dimensional vector

$$\mathbf{b} = \frac{3}{2\pi(L+1) \cdot \mathbf{S}^H \mathbf{P}^{-1} \mathbf{S}} \left[\mathbf{S}_I^T \quad -\mathbf{S}_R^T \quad \mathbf{0}_M^T \quad \frac{L}{\pi(L-1)}\right]^T \tag{47}$$

and observe that multiplying $\mathbf{F}$ by $\mathbf{b}$ results into a vector $\mathbf{e}_{3M+1}$ with elements

$$e_{3M+1}(n) = \begin{cases} 1 & \text{if } n = 3M+1 \\ 0 & \text{otherwise.} \end{cases} \tag{48}$$

This says that $\mathbf{b}$ is actually the last column of $\mathbf{F}^{-1}$ and, in consequence, from (46) we have

$$CRB(\nu) = \frac{3}{2\pi^2 L(1-1/L^2) \cdot \mathbf{S}^H \mathbf{P}^{-1} \mathbf{S}}. \tag{49}$$

Bearing in mind that

$$\mathbf{S}^H \mathbf{P}^{-1} \mathbf{S} = \sum_{m=0}^{M-1} \frac{|S(m)|^2}{\sigma^2(m)} \tag{50}$$

we eventually obtain the results (22) and (23) in the text.

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